

MARLON DE SOUZA

Software Developer - AI Engineer

PERSONAL INFORMATION

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SKILLS

- **Languages:** Advanced Python, SQL, Bash
- **AI & NLP:** LLM APIs, embeddings, Machine Learning and Data processing
- **Vector databases:** Weaviate (admin & schema design, semantic retrieval)
- **Architectures:** RAG, containerized microservices and multi-agent systems.
- **Public Speaking & Mentorship:**
 - Delivered talks on AI & Automation (GDG Joinville, Católica University, internal talks)
 - Mentored developers in AI projects
 - Led internal workshops on Generative AI and best practices

EDUCATION

Católica de Santa Catarina

B.Sc. in Software Engineering
in progress (expected 2025)

PROFILE

AI Engineer and Software Developer with experience in designing and implementing intelligent solutions for real-world business challenges. Skilled in LLMs, RAG, vector databases, and intelligent automations that reduce costs, improve SLA, and increase operational efficiency. Passionate about turning AI into practical results, while contributing through mentoring, talks, and workshops.

WORK EXPERIENCE

Becomex – Software Developer (AI Team)

Oct 2023 – Present | Joinville, Brazil | Hybrid

- Started as an intern in the business team and transitioned to the AI Lab after the success of early AI projects.
- Designed and implemented end-to-end AI applications, from architecture (multi-agent systems, RAG pipelines, vector databases) to production deployment.
- Built the PN Extractor, reducing document review time from 6 hours to 15 minutes (-96% TOIL) with ~98% accuracy.
- Created a full integration pipeline between APIs and internal agents, reducing SLA and operational costs in the tax management line.

KEY PROJECTS

- **Tax Management AI Pipeline:** Designed and deployed a multi-agent, API-integrated pipeline; +70% overall efficiency. Hit 2nd-best SLA in month 1 and later 99.5% SLA after adoption (up from 77–81%).
- **Genetic Mutation Classification (Academic – in progress):** Built training dataset + ML pipeline (FASTA parsing → mutation annotation → MLP/XGBoost/Random Forest) to classify BRCA1/BRCA2 variants (benign vs. pathogenic), optimized for sensitivity.